

ORDEM DE EXECUÇÃO

1) Instalar dependências

➡■ Instala ferramentas necessárias (hardware + Ollama + biblioteca do ngrok).

```
!apt-get install -y zstd lshw pciutils -q
!curl -fsSL https://ollama.com/install.sh | sh
!pip install pyngrok -q
```

2) Verificar GPU

➡■ Confirma se a GPU está disponível.

```
!nvidia-smi
```

3) Subir Ollama e baixar modelo

➡■ Inicia servidor Ollama → baixa modelo → verifica se carregou.

```
import os, subprocess, time

os.system("OLLAMA_HOST=0.0.0.0 CUDA_VISIBLE_DEVICES=0 ollama serve > /tmp/ollama.log 2>&1 &")
time.sleep(10)

subprocess.run(["ollama", "pull", "qwen2.5:14b"], timeout=600)
print("■ Modelo pronto!")

print(subprocess.run(["curl", "-s", "http://localhost:11434/api/tags"],
                    capture_output=True, text=True).stdout)
```

4) Abrir túnel com ngrok

➡■ Cria configuração → expõe Ollama na internet → mostra GPU.

```
import os, subprocess, time

os.makedirs("/root/.config/ngrok", exist_ok=True)

config = """version: "3"
agent:
  authtoken: 3EKvt0o5nbGjVKAfxXMIbNjQx5z_5M53aUu4ox5y7StVVpqHC

tunnels:
  ollama:
    proto: http
    addr: 11434
    domain: semicolon-wasp-user.ngrok-free.dev
    schemes:
      - https
"""

with open("/root/.config/ngrok/ngrok.yml", "w") as f:
    f.write(config)

subprocess.Popen(["ngrok", "start", "ollama", "--log", "/tmp/ngrok.log"])
time.sleep(5)

print("■ Servidor online em: https://semicolon-wasp-user.ngrok-free.dev")
print(subprocess.run(["nvidia-smi"], capture_output=True, text=True).stdout)
```

5) Reiniciar servidor Ollama (se necessário)

➡■ Mata instância antiga → sobe novamente → testa conexão.

```
import os, time, subprocess
```

```
subprocess.run(["pkill", "ollama"], capture_output=True)
time.sleep(3)
```

```
os.system("OLLAMA_HOST=0.0.0.0 CUDA_VISIBLE_DEVICES=0 ollama serve > /tmp/ollama.log 2>&1 &")
time.sleep(8)
```

```
print(subprocess.run(["curl", "-s", "http://localhost:11434"],
capture_output=True, text=True).stdout)
```